

RAHMAH ABDULKARIM

Houston, TX · rahmahabdulkarim@gmail.com · LinkedIn · Portfolio

TECHNICAL SKILLS

Programming & Data: Python, SQL, R, MATLAB, C++, MySQL; data cleaning, relational modeling, pipelines, feature engineering, statistical analysis

Machine Learning: scikit-learn, regression, classification, clustering, dimensionality reduction, model selection, cross-validation, error analysis

Analytics & Engineering: Tableau, Git, Linux, FastAPI, Docker, GitHub Actions, AWS ECR/ECS/Fargate, automated testing, technical documentation

RESEARCH EXPERIENCE

Graduate Research Assistant **Aug 2024–May 2026**
Rice University, in collaboration with Houston Methodist *Houston, TX*

- Developed a reproducible Python workflow linking approximately 18,000 clinical imaging files across AVI, DICOM, and CMR sources; reconciled identifiers, metadata, measurement definitions, and exclusion rules to create analysis-ready cohorts.
- Investigated data, preprocessing, and calibration failures and refined the analytical method, reducing out-of-sample ejection-fraction bias from +5.40 to +0.03 percentage points and mean absolute error from 11.39 to 9.74.
- Communicated methods, assumptions, limitations, and results through documented workflows, figures, presentations, and a master's thesis for clinical and technical collaborators.

REU Research Assistant, Google-Sponsored **May 2023–Aug 2023**
Rice University *Houston, TX*

- Transformed 42,401 irregular geospatial records into 15,345 analytical units with 19 features and compared seven definitions of geographic proximity for seafloor grain-size prediction.
- Designed geographically grouped cross-validation to prevent spatial leakage and selected Extra Trees after it achieved mean RMSE of 2.094, outperforming gradient boosting in every fold by 16.2% on average.

Undergraduate Research Assistant **Feb 2023–Dec 2023**
Howard University *Washington, DC*

- Integrated interview, examination, and laboratory records for 39,156 participants into a quality-controlled cohort of 17,345 adults with 18 demographic, behavioral, and clinical variables.
- Compared five statistical and machine-learning approaches and translated performance, subgroup, and data-quality findings into reproducible tables and figures.

SELECTED PROJECTS

Football Scouting Intelligence System | Python, SQL, MySQL, scikit-learn, FastAPI, Docker, AWS

- Engineered 15 performance features across more than 88,000 player-match observations, producing 1,633 qualified profiles for similarity-based replacement recommendations.
- Evaluated Euclidean, cosine, PCA, and autoencoder representations using neighborhood coherence and split-half and temporal stability; selected Euclidean retrieval after it achieved 66.3% role coherence at 10 versus a 29.6% random baseline.
- Productionized an explainable FastAPI service with 17 automated tests, Docker, GitHub Actions CI, AWS deployment, and a public Streamlit interface.

Football Performance Intelligence | SQL, Python, Tableau

- Queried a normalized 13-table MySQL database with joins, CTEs, aggregations, and window functions to create 11,722 passing records, 4,596 qualified player records, and 312 team profiles; validated results and delivered an interactive Tableau dashboard.

Premier League Manager Impact Analysis | R, Bayesian Statistics

- Modeled 7,600 team-match observations across 10 seasons, controlling for team, opponent, venue, and season effects; identified 10.9% more goals conceded during matches 7–10, with a 95% interval of 3.1%–19.0%.

EDUCATION

Rice University **Houston, TX**
M.A. Computational & Applied Mathematics & Operations Research, GPA: 3.76 *May 2026*

Coursework: Modern Machine Learning, Bayesian Statistics, Stochastic Processes, Advanced Numerical Analysis, High Performance Computing

Howard University **Washington, DC**
B.S. Mathematics; Minors in Computer Science, Chemistry, and Latin, GPA: 3.91 *May 2024*